

RAGHAVA RATHAN APPANA

✉ rathanappana@gmail.com

☎ +46 0734870972

📍 Stockholm, Sweden



Education

KTH Royal Institute of Technology, Stockholm, Sweden

Aug 2022 – Nov 2024

M.Sc. in Information and Network Engineering (Specialized in Networked Systems)

GPA: 3.5/5

- Relevant Courses: Building Networked Systems Security, Wireless Networks, Ethical Hacking

Amrita Vishwa Vidyapeetam, Bangalore, India

Jul 2019 – Jul 2022

B.Tech. in Electronics and Communication Engineering

GPA: 8.78/10

- Relevant Courses: Wireless Communication, Computer Networks, Radio Frequency Engineering

Experience

Visiting Scholar Researcher, KU Leuven, Belgium

Mar 2025 – Aug 2025

- Researching privacy-enhancing mechanisms and secure protocol designs for enterprise network environments.
- Investigating attack vectors and designing countermeasures to strengthen enterprise network security.

Projects

M.Sc Thesis: AP-Initiated Dynamic MAC Address Re-Randomization

Mar 2024 – Nov 2024

- Developed a runtime MAC re-randomization protocol to enhance WiFi privacy and prevent device tracking.
- Implemented kernel-level modifications in mac80211 to enable seamless, dynamic MAC address changes.
- Mininet-WiFi simulations showed MAC linkability dropping from 100% to 15%, significantly reducing tracking risk.
- Ensured minimal network overhead, maintaining stable throughput and latency despite frequent re-randomizations.

Enterprise Network Penetration Testing

Aug 2023 – Oct 2023

- Performed penetration testing and vulnerability exploitation using Metasploit, SQLMap, Hydra, and Burp Suite.
- Gained access and escalated privileges by exploiting SSH, MySQL, Apache Tomcat, and Windows Server.
- Captured security flags through password cracking, RCE, and lateral movement in a corporate cyber range.

Scalable and Secure Network Infrastructure for Distributed Offices

Jan 2023 – Mar 2023

- Designed a secure enterprise network architecture, integrating OpenVPN, firewalls, and reverse proxy controls.
- Implemented an Intrusion Detection System using SNORT, enabling real-time monitoring and anomaly detection.
- Deployed a FreeRADIUS authentication system with certificate-based access control for secure remote connectivity.

B.Tech Thesis: Facial Recognition Based Classroom Attendance System

Feb 2022 – Jul 2022

- Developed an automated attendance system using facial recognition and OpenCV.
- Transitioned project into a product used within the university.

Skills

Programming: C, C++, Python

Networking: Linux, VMware, Bash, Network Protocols, Network Configuration, Network Troubleshooting

Network Security: Firewalls, VPNs, IDS/IPS, Encryption, Penetration Testing, Risk Management

Tools: Wireshark, SSH, Git, Docker, VirtualBox, Cisco Packet Tracer, Mininet-WiFi, Wmediumd

Publications

Measuring and Preventing Certificate Misconfigurations in Enterprise WPA2/3 Networks

- Accepted for publication at CSNet 2025 (Forthcoming).

Achievements

Top 10 Rank in Bachelor of Technology

Jul 2023

- Ranked 10th in the Bachelor of Technology program, Amrita Vishwa Vidyapeetam.